FairCom DB - dbnotify Service Setup Step-by-Step

The 'dbnotify' feature is an optional Plug-in service that is included in the FairCom DB package. This feature allows you to define "event-criteria " on any FairCom DB database file. Using these criteria, the file is actively monitored and a trigger will invoke a 'dbnotify' processing event if/when the criteria are satisfied. Reflecting this database event, a configurable MQTT message is sent to a specified MQTT Broker.

Let's experience this feature in the following steps:

1. Install FairCom DB and FairCom MQ packages: Installation is as easy as unzipping on Windows or untar'ing on Linux/Unix. Create your own folders locations (perhaps DB and MQ) to install these packages:

   > DB Folder: Un-zip (or un-tar) your FairCom DB package:
   FairCom-DB.windows.64bit.vX.x.x.xxx.xxxxxx.ACE.zip

   > MQ Folder: Un-zip (or un-tar) your FairCom MQ package:
   FairCom-MQ.windows.64bit.vX.x.x.xxx.xxxxxx.EDGE.zip

2. In some customer cases, a special FairCom MQ license file has been provided along with your FairCom packages in the form: 'ctsrvr39xxxxxx.broker.lic' Drop this license file into the MQ 'server' folder and rename it to: 'ctsrvr39xxxxxx.lic'

3. Check Ports on the Broker: Because we are running the MQ Broker and the DB Server on the same machine, we need to change a few port numbers so they will not conflict.

   Later we will use SQL, so we should adjust our SQL port on our MQ Broker to be different from the DB Server. Edit your MQ Broker's configuration file: ./config/ctsrvr.cfg. Un-comment the SQL_PORT (by removing the semi-colon) and change the number to 6598.

   MQ: Edit this file: .\config\ctsrvr.cfg

   Change this:
   ```
   ;Default SQL port
   ;SQL_PORT          6597
   ```

   to this:

   ```
   ; Default SQL port
   SQL_PORT           6598
   ```

Next, we do not want the HTTPS Server's ports to conflict.

MQ: Edit the MQ Broker's '.\config\services.json' file.

For the "protocol": "https", change 8443 to 8444, like this.

```
"serviceName": "https8444",
"description": "Port 8444 using TLS-secured HTTPS protocol for REST and Web Apps on all TCP/IP addresses bound to this server",
"port": 8444,
"protocol": "https",
"enabled": true,
```

4. Start your Broker: From within your MQ 'server' folder, click on the 'faircom.exe' to start your MQTT Broker.

5. Below, you will see that we are setting up the 'dbnotify' service to monitor a file named 'custmast' and sending MQTT messages to a topic named "topic1' on the FairCom MQ Broker. Our 'topic1' needs to exist on the MQ Broker for 'dbnotify' to work.

   There are two ways that 'topic1' gets created:

   a. When x509 is NOT activated, 'topic1' will be created automatically by 'dbnotify'.

   b. If x509 is activated, 'topic1' must be created manually.

   See the end this document for manual instructions:

"FairCom DB - dbNotify Service - Secure Communications to MQ Broker with SSL"

As we have not activated SSL yet, we are OK to move forward.

6. Start your DB Server: From within your DB 'server' folder, click on the 'faircom.exe' to start your FairCom DB Server.

7. Create your 'custmast' file: Below we will be configuring the 'dbnotify' service to monitor our database file named 'custmast'. We have a current dependency that the file being monitored must exist when the Server is started and the 'dbnotify' service is initialized. Before we configure our 'dbnotify', let's go ahead and create our 'custmast' file so we will be ready to go.

   The 'isql' command line utility is located in the 'tools' folder. We include two SQL scripts. Copy both of them to your ./tools folder.

   ```
   copy \config\x509.auth.demo\custmast1.sql .\tools
   copy \config\x509.auth.demo\custmast2.sql .\tools
   ```

Now execute 'custmast1.sql' to create our 'custmast' file.

isql -s custmast1.sql -u ADMIN -a ADMIN ctreeSQL

See:
```
\FairCom-DB.windows.64bit.vXX.X.X.XX\tools>isql -s custmast1.sql -u ADMIN -a
ADMIN ctreeSQL
INIT
DEFINE
Create table...
MANAGE
Add records...
1 record inserted.
1 record inserted.
1 record inserted.
1 record inserted.
Display records...
NUMBER NAME
------ ----
1000   Bryan Williams
1001   Michael Jordan
1002   Joshua Brown
1003   Keyon Dooling
4 records selected
DONE
```

You can see your files on disc here:
\FairCom-DB.windows.64bit.vXX.X.X.XX\data\ctreeSQL.dbs

admin_custmast.dat
admin_custmast.idx

8. Shutdown your DB Server: With our 'custmast' created, we now proceed to configure the 'dbnotify' and later restart the DB server with our new configuration.

9. Activate the 'dbnotify' Service: Before you restart your DB Server, activate the 'dbnotify' service for your FairCom DB server. Open the 'services.json' configuration file in the 'config' folder where you installed the DB product.

   Search for 'dbnotify' and enable this feature by setting "enabled": true

   DB: Edit .\config\services.json
   Set "enabled": true

```
"otherServices": [
{
        "serviceName": "dbnotify",
```

```
        "serviceLibrary": "./dbnotify/fcdbnotify.dll",
        "enabled": true    <<<<<< HERE <<<<<
},
```

10. Configure your DB Connection to the MQTT Broker: See the 'dbnotifyconnections.json' file in your DB 'config' folder: The 'faircomServerName' should be set to the 'SERVER_NAME' from your MQ broker's './config/ctsrvr.cfg' file.

    As you can see, our default is set like this: 'SERVER_NAME FCEDGEMQ' Therefore, it looks like the 'faircomServerName' in our DB's 'dbnotifyconnections.json' is OK!

    We will look at other properties later in this tutorial when we explore security.

    DB: See: .\config\dbnotifyconnections.json"

```
{
  "directConnectionsToFairComMq":
  [
    {
      "brokerConnectionName": "brokerCtree",
      "faircomServerName": "FCEDGEMQ",                    <<<<< HERE <<<<<
      "brokerHostname": "localhost",
      "brokerUserName": "ADMIN",
      "brokerUserPassword": "ADMIN",
      "passwordCertificateFilename": "",
      "reconnectFrequencySeconds": 15,
      "tls":
      {
        "certificateFilename": "ctsrvr.pem",
        "enabled": false,

        "x509_authentication":
        {
          "certificateFilename": "admin.pem",
          "privateKeyFilename": "admin.key",
          "enabled": false
        }
      },
      "metadata": {},
      "defaultIncludePrimaryKey": "never",
      "defaultTagChanges":       "neverTag",
    }
  ]
}
```

11. Configure your DB notification criteria: When the DB Server enables the 'dbnotify' service at startup, it looks in the 'config\dbnotify' folder for any .json files. Each .json file is expected to contain the necessary information related to one database file to

be monitored. You need a separate .json file for each database file you want to monitor.

We have provided an example .json file for this tutorial.

DB: See '.\config\dbnotify\ctreeSQLcustmast.json'

```
{
   "databaseName": "ctreeSQL",
   "tableName": "custmast",
   "ownerName": "admin",

   "publishMqttMessages": [
      {
         "topic": "topic1",
            "persistenceDatabaseName":"faircom",
            "persistenceOwnerName":"admin",
            "persistenceTableName":"mqtt_msg_topic1",
         "brokerConnectionName": "brokerCtree",
         "triggers": [ "delete"],
         "QoS": 0,
         "recordFilter": "!stricmp(cm_custstat,\"CA\")",
         "includedFields": [ "cm_custnumb", "cm_custzipc", "cm_custcity" ],
         "includePrimaryKey": "never"
      },
      {
         "topic": "topic1",
            "persistenceDatabaseName":"faircom",
            "persistenceOwnerName":"admin",
            "persistenceTableName":"mqtt_msg_topic1",
         "brokerConnectionName": "brokerCtree",
         "triggers": [ "insert", "update"],
         "QoS": 0,
         "recordFilter": "atoi(cm_custnumb) >= 1000",
         "includePrimaryKey": "never",
         "tagChanges":       "tagEachField"
      }
   ]
}
```

- The 'custmast1.sql' SQL script we ran above uses the 'ctreeSQL' database name.
- It creates a file named 'custmast' with owner 'admin'.
- We will publish to "topic": "topic1" on "brokerConnectionName":"brokerCtree".

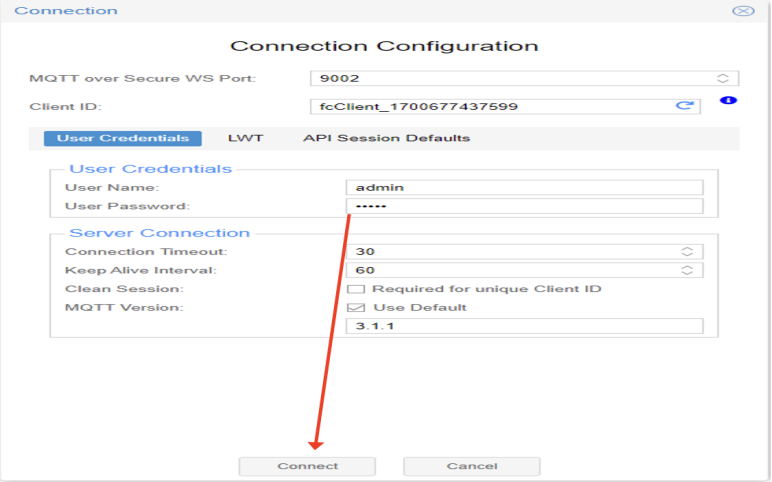
Leave all these settings intact (out of the box) for this demo.

12. Start your DB Server: From within your DB 'server' folder, click on the 'faircom.exe' to re-start your DB Server.

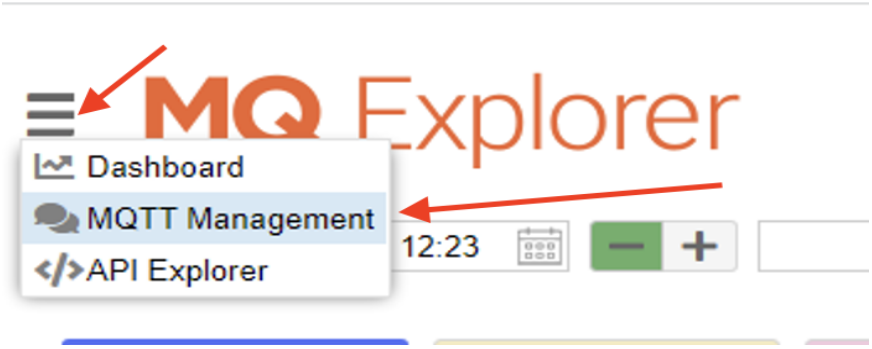
13. MQExplorer: Subscribe to "topic1" in the MQExplorer. From your Browser, load MQExplorer:

    https://localhost:8444/MQExplorer

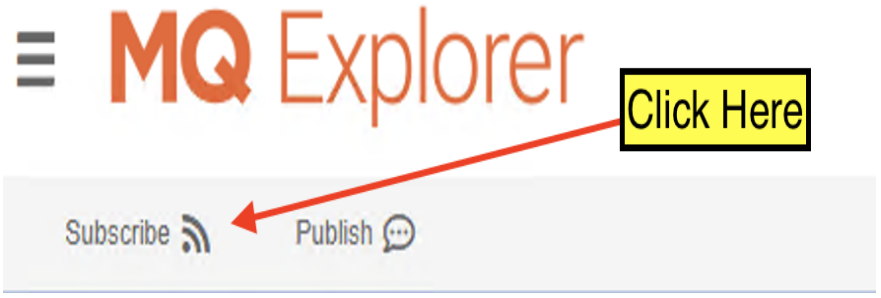
    Connect with the default credentials.



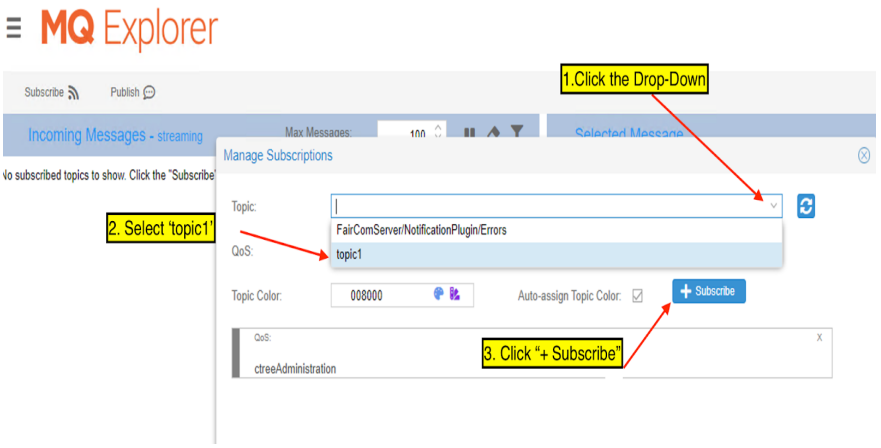
- Open "MQTT Management" from the top-left corner menu.



- Click at "Subscribe".



Select "topic1". Click at "+ Subscribe" and close this window.



Hit Close.

Leave this window open to see "Incoming Messages".

14. Instigate some 'dbnotify' activity and view MQTT messages in MQExplorer:

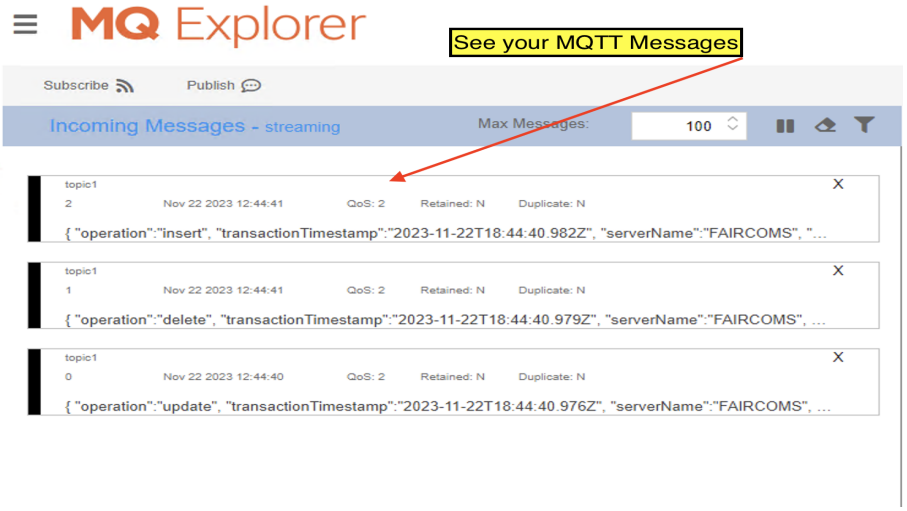
    DB: Return to your FairCom DB './tools' folder and execute 'custmast2.sql'. This will trigger a 'dbnotify' event, and then you can inspect your MQTT Broker for messages sent to 'topic1'.

    From your DB ./tools folder, execute

    isql -s custmast2.sql -u ADMIN -a ADMIN ctreeSQL

    This script executes an update, a delete and an insert command for 'custmast' that will instigate notifications.

Return to your MQExplorer to see the messages. It is fine to run 'custmast2.sql' over and over again to see more incoming messages into MQExplorer.



OK, we are good. This activity from DB Server is now reflected in the MQ Broker via MQTT messages.

END OF DOC